VQE

The Variational Quantum Eigensolver

A step-by-step chapter: from eigenvalue problems to a CUDA-Q implementation

What you will be able to do after this chapter

- solve small eigenvalue problems, compute Rayleigh quotients and apply the variational principle;
- set up an electronic-structure problem: Born–Oppenheimer, orbitals, determinants, correlation energy;
- use second quantisation and map fermions to qubits with Jordan–Wigner and Bravyi–Kitaev;
- decompose Hamiltonians into Pauli strings and compute expectation values from measurements;
- design and analyse ansätze (Hartree–Fock, UCC, hardware-efficient) and classical optimisation loops;
- read and modify `VQE.ipynb`, and interpret its results, shot requirements and noise honestly;
- solve every type of question in the accompanying 80-question VQE assessment.

Companion material in the same repository: `presentations/VQE_Lecture.pdf` (slides), `VQE.ipynb` (CUDA-Q notebook) and `assessments/` (question paper and answer book).

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VQE — The Variational Quantum Eigensolver

How to use this chapter

Sections 1–3 are the *mathematics and chemistry* (eigenproblems, molecular Hamiltonians, second quantisation). Sections 4–5 are the *quantum-computing toolkit* (fermion-to-qubit mappings, Pauli strings, measurement). Sections 6–9 are *VQE itself* (variational principle, ansätze, optimisation, worked examples). Sections 10–12 cover the *implementation*, shots, noise and limitations. Section 13 maps every assessment question to the section that teaches it; Section 14 has practice problems with answers. Statements are marked as general theory, the notebook's implementation, or illustrative examples; "previously recorded" results are the outputs stored in `VQE.ipynb`.

Definition: Conventions used everywhere in this chapter

- Pauli words: character k acts on qubit k; basis states $|q_0 q_1 \ldots\rangle$ are written qubit 0 first.
- $R_Y(\theta) = e^{-i\theta Y/2}$; CUDA-Q's `exp_pauli(theta, q, word)` $= e^{+i\theta P}$.
- Jordan–Wigner with $|1\rangle$ = occupied: $a_j^\dagger = Z_0 \cdots Z_{j-1} \frac{1}{2}(X_j - iY_j)$.
- Energies in Hartree (Ha); 1 Ha = 627.51 kcal/mol; chemical accuracy 1.6 mHa.

1 Eigenvalue problems

A Hermitian matrix H has real eigenvalues, an orthonormal eigenbasis, and the spectral decomposition $H = \sum_k E_k |E_k\rangle \langle E_k|$. The **ground-state energy** is the smallest eigenvalue E_0.

Key result: Rayleigh quotient

For every non-zero v: $E_0 \le R(v) = \dfrac{v^\dagger H v}{v^\dagger v} \le E_{\max}$, with equality at eigenvectors. If $\|v\| = 1$, $R(v) = \langle v|H|v\rangle$.

Worked example 1.1: a 2×2 problem

$H = \left(\begin{smallmatrix} 1 & 2 \\ 2 & -2 \end{smallmatrix}\right)$: $\lambda^2 + \lambda - 6 = 0$, $\lambda \in \{-3, 2\}$; ground vector $(1, -2)/\sqrt{5}$. For $v = (1, -1)$: $Hv = (-1, 4)$, $R = -5/2 = -2.5 \ge -3$. Weight of the ground vector in $v/\|v\|$: 0.9; check $0.9(-3) + 0.1(2) = -2.5$.

Common trap

The ground energy lies *below* the smallest diagonal element when off-diagonal couplings exist ($-3 < -2$). And: trace = sum, determinant = product of eigenvalues — use them to check answers.

Classically, dense diagonalisation costs $O(D^3)$ and sparse Lanczos methods $O(D)$ per iteration, with $D = 2^n$ for n qubits. The notebook diagonalises up to 18 qubits (`REFERENCE_MAX_QUBITS`) and solves the Ising chain exactly for any size via free fermions.

2 Quantum chemistry essentials

2.1 Schrödinger equation and Born–Oppenheimer

$i\hbar\partial_t |\psi\rangle = H|\psi\rangle$; eigenstates evolve only by a global phase $e^{-iEt/\hbar}$ (stationary states), so energies come from $H|\psi\rangle = E|\psi\rangle$. With nuclei fixed (Born–Oppenheimer), the nuclear repulsion V_{NN} is a constant that ends up in the identity coefficient of a qubit Hamiltonian.

2.2 Orbitals, determinants, correlation

A basis of N_{orb} spatial orbitals gives $N = 2N_{\text{orb}}$ spin-orbitals; a Slater determinant is a choice of N_e occupied spin-orbitals: $\binom{N}{N_e}$ of them. Hartree–Fock (HF) is the best single determinant; full configuration interaction (FCI) diagonalises over all of them. $E_{\text{corr}} = E_{\text{FCI}} - E_{\text{HF}} \leq 0$.

Worked example 2.2: H_2 in STO-3G (the notebook's molecule)

2 spatial orbitals (σ_g, σ_u), 4 spin-orbitals, 2 electrons: $\binom{4}{2} = 6$ determinants, 4 with $S_z = 0$. $E_{\text{HF}} = -1.116759$ Ha, $E_{\text{FCI}} = -1.137284$ Ha: $E_{\text{corr}} = -20.5$ mHa ≈ 13 kcal/mol, far above chemical accuracy.

Worked example 2.3: LiH (illustration, not in the notebook)

STO-3G: Li $1s, 2s, 2p_{x,y,z}$ + H $1s$ = 6 spatial orbitals → 12 qubits, 4 electrons; PySCF + OpenFermion give 631 Pauli terms. Freezing the Li $1s$ core removes 2 qubits.

Common trap

At large bond length, restricted HF puts 50 % weight on ionic configurations and lies far too high (static correlation). The nuclear repulsion has nothing to do with this error: it is identical in every method.

3 Second quantisation

$\{a_p, a_q^\dagger\} = \delta_{pq}$, $\{a_p, a_q\} = 0$; hence $(a_p^\dagger)^2 = 0$ (Pauli exclusion) and $n_p = a_p^\dagger a_p$ has eigenvalues 0, 1.

$$H = \sum_{pq} h_{pq} a_p^\dagger a_q + \tfrac{1}{2} \sum_{pqrs} h_{pqrs} a_p^\dagger a_q^\dagger a_r a_s + E_{\text{const}}$$

has $O(N^4)$ terms. The integrals come from a classical code (PySCF); the quantum computer only has to handle the many-electron state.

4 From fermions to qubits

Key result: Jordan–Wigner

$a_j^\dagger = Z_0 \cdots Z_{j-1} \frac{1}{2}(X_j - iY_j)$, $n_j = \frac{1}{2}(I - Z_j)$. Hopping: $a_j^\dagger a_{j+1} + \text{h.c.} = \frac{1}{2}(X_jX_{j+1} + Y_jY_{j+1})$. A double excitation $a_3^\dagger a_2^\dagger a_1 a_0 - \text{h.c.}$ becomes 8 Pauli strings with coefficients $\pm i/8$.

Worked example 4.4: hopping, derived

$a_0^\dagger a_1 = \frac{1}{2}(X_0 - iY_0) \cdot Z_0 \frac{1}{2}(X_1 + iY_1)$ and $(X - iY)Z = X - iY$, so $a_0^\dagger a_1 = \frac{1}{4}(X_0 - iY_0)(X_1 + iY_1)$. Adding the Hermitian conjugate cancels the cross terms: $\frac{1}{2}(X_0X_1 + Y_0Y_1)$.

Other encodings. Parity mapping (qubit j stores $n_0 \oplus \cdots \oplus n_j$); Bravyi–Kitaev ($b = Bn \bmod 2$ on a binary tree, Pauli weight $O(\log N)$; for 4 modes the HF occupation $(1, 1, 0, 0)$ becomes $|1000\rangle$); tapering removes qubits using $\mathbb{Z}_2$ symmetries.

Common trap

The same physical state has different bitstrings in different encodings: the HF state is $|1100\rangle$ under JW but $|1000\rangle$ under BK.

The notebook's H_2 Hamiltonian has 15 Jordan–Wigner terms (identity, 10 Z-type, 4 exchange terms `XXYY`, `XYYX`, `YXXY`, `YYXX`). PySCF + OpenFermion reproduce every coefficient to 10^{-15} at $R = 0.740$ Å (the notebook's text says 0.7414 Å, where the coefficients differ by up to 1.8 mHa).

5 Pauli strings and measurement

- 4^n Pauli strings form an orthogonal basis: $c_P = 2^{-n} \operatorname{tr}(PH)$, real for Hermitian H. Example: $\left(\begin{smallmatrix} 3 & 1-2i \\ 1+2i & -1 \end{smallmatrix}\right) = I + X + 2Y + 2Z$; $\operatorname{diag}(1, 2, 3, 4) = 2.5\, II - ZI - 0.5\, IZ$.
- Pauli algebra: $XY = iZ$ (cyclic), strings multiply position by position; they commute iff they anticommute at an even number of positions.
- $e^{i\theta P} = \cos\theta + i \sin\theta P$; on an eigenvector $e^{i\theta P} |v\rangle = e^{i\theta\lambda} |v\rangle$. $X_0X_1X_2Y_3 |1100\rangle = i |0011\rangle$.
- Measuring: $X \to H$, $Y \to S^\dagger$ then H, $Z \to$ nothing; outcome $(-1)^{\text{parity of measured bits}}$; standard error $\leq 1/\sqrt{S}$.
- Grouping: qubit-wise commuting strings share single-qubit basis changes. H_2: 5 QWC settings, 2 with general commuting groups.
- Ordering: CUDA-Q state vectors are little-endian (qubit 0 least significant): $|1100\rangle$ has index 3.

Key result: zero-variance principle

$\operatorname{Var}(H) = \langle H^2 \rangle - \langle H \rangle^2 = 0$ exactly for eigenstates; it sets the shot noise $\sqrt{\operatorname{Var}/S}$ of energy estimates.

6 The variational principle and the VQE loop

Key result: variational principle

For every normalised $|\psi\rangle$ (and every density matrix ρ): $\langle\psi|H|\psi\rangle = \sum_k |c_k|^2 E_k \geq E_0$, $\mathrm{tr}(\rho H) \geq E_0$. Rayleigh–Ritz: Ritz values in any subspace are upper bounds to the corresponding exact eigenvalues.

VQE: prepare $|\psi(\boldsymbol{\theta})\rangle = U(\boldsymbol{\theta})\,|0\ldots0\rangle$ (normalised because gates are unitary), estimate $E(\boldsymbol{\theta}) = \sum_k c_k \langle P_k\rangle$, update $\boldsymbol{\theta}$ classically, repeat; report the lowest energy found.

Common trap

An energy below E_0 computed with *exact* expectation values is a bug (Hamiltonian, reference, normalisation). With finite shots, estimates can fall below E_0; error-mitigated (extrapolated) values have no variational guarantee either.

Worked example 6.5: one qubit

$H = -Z + \frac{1}{2}X$, $|\psi(\theta)\rangle = R_Y(\theta)\,|0\rangle$: $\langle Z\rangle = \cos\theta$, $\langle X\rangle = \sin\theta$, $E(\theta) = -\cos\theta + \frac{1}{2}\sin\theta$. $E' = 0 \Rightarrow \tan\theta = -\frac{1}{2}$, $\theta^* = -0.4636$, $E(\theta^*) = -\sqrt{5}/2 = -1.118 = E_0$. Start $\theta = 0$: error 0.118, variance 0.25.

7 Ansatz design

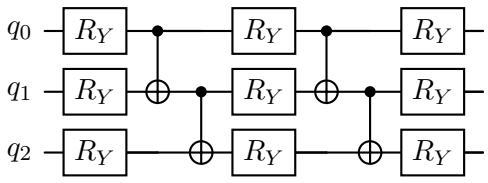
7.1 Hartree–Fock and UCC

HF under JW is a basis state (H_2: $|1100\rangle$, two X gates; energy from the identity and Z terms only, -1.116759 Ha). UCC: $|\psi\rangle = e^{T-T^\dagger}\,|\mathrm{HF}\rangle$ with singles and doubles; counts $n_o n_v$ and $\binom{n_o}{2}\binom{n_v}{2}$ (LiH: 32 and 168; H_2: 4 and 1).

Key result: the notebook's UCC for H_2

$|\psi(\theta)\rangle = e^{i\theta X_0X_1X_2Y_3}\,|1100\rangle = \cos\theta\,|1100\rangle - \sin\theta\,|0011\rangle$. The exact ground state lies in this plane, so one parameter reaches FCI: $\theta^* = 0.112783$, $E = -1.137283834$ Ha (CUDA-Q and NumPy agree to 10^{-15}).

7.2 Hardware-efficient ansatz



L layers of [R_Y on all qubits, CNOT ladder] + final R_Y: $(L+1)N$ parameters, $L(N-1)$ CNOTs. Default $N = 15$, $L = 2$: 45, 28, 73 gates. Real gates only: fine for real Hamiltonians (Ising, H_2). Initial angles $\mathcal{N}(0, 0.1^2)$, i.e. near $|0\ldots0\rangle$ (Ising energy -14).

Common trap

Deep random hardware-efficient circuits suffer from barren plateaus (gradients vanishing exponentially in n); chemistry-inspired ansätze, HF starts and local cost functions help.

8 Classical optimisation

COBYLA (trust region, derivative-free; notebook default with `rhobeg = 0.3`, `maxiter = 300`), Nelder–Mead (simplex), Powell (conjugate directions), L-BFGS-B (quasi-Newton; finite-difference gradients, $n_p + 1$ runs). Exact parameter-shift gradients for gates $e^{-i\theta P/2}$: $\partial_\theta E = \frac{1}{2}[E(\theta + \frac{\pi}{2}) - E(\theta - \frac{\pi}{2})]$, $2n_p$ runs. The notebook keeps the *best* evaluated angles (`argmin(energy_history)`), not necessarily `opt.x`. A `MAXFUN` message means the budget ended the run.

9 Worked examples

Worked example 9.6: H_2 as a 2×2 problem

Step 1. $H_{11} = \langle 1100|H|1100\rangle = -1.116759$, $H_{22} = \langle 0011|H|0011\rangle = 0.462618$, $H_{12} = 0.181210$ Ha (from the 15 terms).

Step 2. $E(\theta) = H_{11}\cos^2\theta + H_{22}\sin^2\theta - H_{12}\sin 2\theta$.

Step 3. $\tan 2\theta^* = 2H_{12}/(H_{22} - H_{11}) = 0.2295$: $\theta^* = 0.1128$; $P(|0011\rangle) = \sin^2\theta^* = 0.0127$.

Step 4. $E_{\min} = \frac{H_{11}+H_{22}}{2} - \sqrt{(\frac{H_{22}-H_{11}}{2})^2 + H_{12}^2} = -1.137284$ Ha $= E_{\text{FCI}}$.

Worked example 9.7: two-qubit Ising chain

$H = -Z_0Z_1 - \frac{1}{2}(X_0 + X_1)$. Circuit $R_Y(\pi/2) \otimes I$, CNOT: Bell state, $E = -1$. Exact: block $\{|\Phi^+\rangle, |\Psi^+\rangle\}$ with $\begin{pmatrix} -1 & -1 \\ -1 & 1 \end{pmatrix}$, $E_0 = -\sqrt{2}$; relative error 29.3 %. After the CNOT $\langle Z_0Z_1\rangle = \cos\theta_1$ in general.

10 The implementation in `VQE.ipynb`

- Hamiltonians: `"tfim"` ($-J\sum Z_iZ_{i+1} - h\sum X_i$, default $N = 15$, 29 terms), `"H2"` (15 terms), `"custom"` (real coefficients only), built with `cudaq.SpinOperator.from_word`.
- Exact references: free fermions ($2N \times 2N$ matrix, any N) for TFIM, cross-checked by sparse diagonalisation for $N \leq 16$; sparse diagonalisation up to 18 qubits otherwise.
- Kernels: `hea_ansatz`, `ucc_ansatz` (+ sampled versions); energy: `cudaq.observe(kernel, hamiltonian, *args).expectation()`; optimiser: `scipy.optimize.minimize`; final `cudaq.sample` with 4000 shots.
- Backends: `qpp-cpu`, `density-matrix-cpu`, `nvidia`, `tensornet(-mps)`, multi-GPU via MPI, hardware providers; `"gpu"` without a GPU raises `RuntimeError`. Only the CPU path has been executed.
- Reports and exports in `exports/vqe/`: circuit drawing, QIR, JSON/PNG report.

Previously recorded default run: $E_{\mathrm{VQE}} = -15.0533898284$ vs. $E_0 = -15.0825179883$ (error 0.0291, 0.193 %), 300 evaluations (budget), 69.10 s quantum execution (0.23 s per evaluation), $|0\ldots0\rangle$ sampled with probability 0.70 (symmetry breaking). Measured from the product state (-14), VQE recovered 97.3 % of the remaining energy. The lecture's own run and a re-execution of the notebook reproduce these numbers on a different CPU (see the lecture).

11 Shots, noise and mitigation

- Shots: $S = (\sigma/\varepsilon)^2$; $\sigma = 0.1$ Ha, $\varepsilon = 1.6$ mHa $\Rightarrow$ 3906 shots per energy evaluation.
- Complete depolarisation: $E = \mathrm{tr}(H)/2^n = c_{I\ldots I}$ (-0.0971 Ha for H_2); noise raises energies but keeps $\mathrm{tr}\,\rho H \geq E_0$.
- Zero-noise extrapolation: $E(1) = -1.10$, $E(3) = -1.00 \Rightarrow E(0) = -1.15$ (linear); no variational guarantee.
- Memory: state vector $16 \cdot 2^n$ B (12 qubits: 65.5 kB), density matrix $16 \cdot 4^n$ B (12 qubits: 268 MB).

12 Limitations and advanced methods

Measurement cost ($O(N^4)$ terms $\times$ shots $\times$ evaluations), barren plateaus, local minima, limited expressiveness and noise limit VQE; no chemical quantum advantage has been demonstrated. Advanced directions: ADAPT-VQE, k-UpCCGSD, active spaces and tapering, excited-state methods (variational quantum deflation, subspace-search VQE, quantum subspace expansion), error mitigation. Applications: reaction energies, potential surfaces, materials and magnetism.

13 Problem-solving strategies and question map

Method: five habits

1. Fix the conventions: qubit order, occupied = $|1\rangle$, sign of `exp_pauli`.
2. Diagonal terms first: on a basis state only I/Z strings contribute.
3. Use symmetry to shrink matrices (2×2 blocks).
4. Check every energy against the variational bound.
5. Separate recorded, re-executed and illustrative numbers.

Section	Assessment questions
1 Eigenvalue problems	1, 2, 3, 8, 13, 14, 15
2 Quantum chemistry	4, 5, 6, 9, 10, 18, 19, 20, 77, 78
3 Second quantisation	11, 12
4 Fermions to qubits	61, 62, 63, 64, 65, 67
5 Pauli strings, measurement	16, 17, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 38, 39, 40, 47, 54
6 Variational principle	7, 37, 41, 42, 53, 71
7 Ansatz design	43, 44, 49, 56, 66
8 Optimisation	51, 52, 58, 68
9 Worked examples	45, 48, 59, 60
10 Implementation	46, 50, 55, 57, 73, 74, 75, 76, 79, 80
11 Shots, noise	69, 70, 72

14 Practice problems

P1. Eigenvalues of $\left(\begin{smallmatrix} 2 & 1 \\ 1 & 2 \end{smallmatrix}\right)$ and the Rayleigh quotient of $(1, 0)$.

P2. Pauli decomposition of $\left(\begin{smallmatrix} 0 & -i \\ i & 2 \end{smallmatrix}\right)$.

P3. Number of 2-electron determinants in 6 spin-orbitals.

P4. Jordan–Wigner form of $n_0 n_1$.

P5. $\langle \Phi^+ | X_0 X_1 | \Phi^+ \rangle$ and $\langle \Phi^+ | Z_0 | \Phi^+ \rangle$ for the Bell state.

P6. For $H = 2Z + X$, trial $R_Y(\theta)\,|0\rangle$: $E(\theta)$ and E_0.

P7. HEA counts for $N = 8$, $L = 4$.

P8. $E(\theta)$ for $\cos\theta\,|u\rangle - \sin\theta\,|v\rangle$ with $H_{uu} = 0$, $H_{vv} = 1$, $H_{uv} = -0.5$; its minimum.

P9. Shots for 0.5 mHa precision with $\sigma = 0.05$ Ha.

P10. Energy of $|0 \ldots 0\rangle$ for the Ising chain with $N = 10$, $J = 2$, $h = 0.3$.

Answers to the practice problems

P1. 1 and 3; $R((1,0)) = 2$.

P2. $c_I = 1$, $c_Z = -1$, $c_X = 0$, $c_Y = 1$: $I + Y - Z$ (check $M_{01} = c_X - ic_Y = -i$).

P3. $\binom{6}{2} = 15$.

P4. $n_0 n_1 = \frac{1}{4}(I - Z_0)(I - Z_1) = \frac{1}{4}(I - Z_0 - Z_1 + Z_0 Z_1)$.

P5. 1 and 0.

P6. $E = 2\cos\theta + \sin\theta$; $E_0 = -\sqrt{5}$.

P7. 40 parameters, 28 CNOTs, 68 gates.

P8. $E = \sin^2\theta + 0.5\sin 2\theta$; eigenvalues of $\left(\begin{smallmatrix} 0 & -0.5 \\ -0.5 & 1 \end{smallmatrix}\right)$: $\frac{1}{2} \pm \frac{\sqrt{2}}{2}$, minimum -0.2071.

P9. $(0.05/0.0005)^2 = 10\,000$.

P10. $-2 \times 9 = -18$ ($\langle X_i \rangle = 0$).

Summary in ten lines

1. Ground energy = smallest eigenvalue of a Hermitian H; Rayleigh quotients bound it from above.
2. Born–Oppenheimer + basis → second-quantised H with $O(N^4)$ terms; constants (incl. V_{NN}) sit in $c_{I\ldots I}$.
3. Jordan–Wigner: occupation per qubit + Z-strings; BK: $O(\log N)$ weight.
4. $c_P = 2^{-n}\,\mathrm{tr}(PH)$; measure P after H / $S^\dagger H$ rotations; group commuting terms.
5. Variational principle: $\langle\psi|H|\psi\rangle \geq E_0$ for every normalised (or mixed) state.
6. VQE = circuit state + measured energy + classical optimiser.
7. H_2: HF $|1100\rangle$ + $e^{i\theta XXXY}$ gives FCI exactly ($\theta^* = 0.1128$).
8. Hardware-efficient: $(L+1)N$ angles, $L(N-1)$ CNOTs; trainability issues when deep.
9. Shots scale as $(\sigma/\varepsilon)^2$; noise pulls energies towards $c_{I\ldots I}$.
10. Read results honestly: exact vs. shot energies, budget-limited runs, recorded vs. re-executed.